

BRAD CAO

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EDUCATION

Columbia University

B.A. in Mathematics & Computer Science

New York, NY

Expected May 2029

Relevant Coursework: Data Structures in Java, Probability Theory, Natural Language Processing, Honors Mathematics A & B

EXPERIENCE

Research Fellow

Sep. 2026 – Present

SPAR (Supervised Program for Alignment Research)

Remote

- Building datasets that evaluate human+AI judge systems for scalable oversight with Rishub Jain and Joshua Jacob (Sampura Research): a 100+ dataset leaderboard spanning deception, jailbreak robustness, cultural bias, and computer-control mistakes
- Evaluating existing judge datasets for onboarding and creating new ones for taxonomy gaps, with sandwiching for ground truth; measuring judge robustness to reward hacking under adversarial RL optimization

Data Science Intern

Apr. 2026 – Present

Mercor

San Francisco, CA

- Engineered evaluation frameworks that formalize subject-matter expertise into RLHF-compatible annotation schemas for LLM fine-tuning pipelines
- Designed rubric-based scoring taxonomies and QA workflows to raise inter-annotator agreement; audited quality and flagged edge cases to refine labeling guidelines

Research Assistant

Apr. 2026 – Present

Duke University

Durham, NC

- Trained and developed LLMs to identify and quantify inter-network disparagement in televised news media, building NLP pipelines to ingest, segment, and label large transcript volumes
- Benchmarked prompt-engineered and fine-tuned classifiers against human-coded samples with Dr. Philip Napoli, translating media-studies frameworks into quantifiable schemas

Research Intern

Jun. 2025 – Aug. 2025

University of Notre Dame

South Bend, IN

- Awarded a full-ride scholarship for original quantitative research on the pharmaceutical industry's impact on public health policy
- Applied R-based regression and visualization to link industry practices to poverty outcomes; presented at a symposium

Research Intern

Jun. 2023 – Sep. 2025

University of Maryland, Baltimore

Baltimore, MD

- First-authored a supervised ML model with Prof. Allison Dunn predicting ADHD-related patient safety risks from real-world data
- First-authored a study modeling tacrolimus dose requirements and transplant outcomes in lung transplant recipients on systemic azole antifungals
- Presented at the ASCPT Annual Meetings (2024, 2025); earned the Top Poster Award (May 2025)

PROJECTS

Threshold | *Python, NumPy, SciPy, PyTorch* | github.com/bradca0/threshold

- Reproduced double descent on MNIST with a random-feature model (nested random ReLU features, min-norm least squares via SVD/Gram decompositions in float64; $n = 2,000$, 5 seeds): test error peaks at $p/n = 1$ (88.8% vs. 24.3% at $p = 5n$; 20% label noise)
- Explained the peak: $\|\beta\|_2$ and the smallest singular value of Φ are extreme at $p = n$ ($\|\beta\|_2$: 811 vs. 2.6 at $p = 5n$); a bias-variance split over 10 draws puts 90% of peak error in variance; tuned ridge cuts the peak from 88.8% to 12.3%
- Showed more data can hurt (fixed $p = 2,000$: test error 34.8% $\rightarrow$ 89.1% as n grows 250 $\rightarrow$ 2,000) and replicated peak and mechanism on Fashion-MNIST; a width-swept MLP showed no peak, reported as a negative result; 85 tests, 94% coverage, `mypy --strict`

MyCureHope | *Django, PostgreSQL, JavaScript, TailwindCSS* | mycurehope.org

- Co-developed an open-source patient database of treatments, doctors, and hospitals serving 33,000+ users across 78 countries; own search build and technical development
- Built and deployed the full-stack platform (Django, PostgreSQL, JavaScript, TailwindCSS) on Nginx/Gunicorn/Ubuntu

PUBLICATIONS

Cao, B., Lee, S., Naga, A., Gobburu, J., Dunn, A. "Leveraging Supervised Machine Learning to Identify Predictors of Serious Clinical Outcome with Stimulant Treatment from Real-World Data." *ASCPT Annual Meeting*, 2025.

Cao, B., Gupta, P., Dunn, A. "Predictive Modeling of Tacrolimus Dose Requirements and Transplantation Outcomes in Lung Transplant Recipients on Systemic Azole Antifungals." *ASCPT Annual Meeting*, 2024.

TECHNICAL SKILLS & INTERESTS

Languages: Python, JavaScript, Java, SQL (PostgreSQL), R, HTML/CSS

Frameworks & Tools: PyTorch, TransformerLens, NumPy, pandas, scikit-learn, Django, TailwindCSS, Nginx, Gunicorn, Ubuntu/Linux

ML & Data: Mechanistic interpretability, LLM fine-tuning, RLHF annotation, supervised ML, data analysis

Interests: Fencing (USA Fencing First Team All-American), Competitive Programming (USACO), History Olympiads, Open-Source